TEAM HACKTIONERS · HACKATHON PROBLEM 1C · SPACE DEBRIS RISK PREDICTION WITH AI

OrbitGuard AI

Our pitch, explained simply — and fact-checked

No space background needed. Page 2 explains the whole project in one minute, page 3 explains every technical word, and then each slide is explained in plain English with proof for every claim.

23	8	7	2	1
TRUE	PROBABLY TRUE	REWORD	NOT PROVEN YET	WRONG

41 claims checked across 6 slides.

8 October 2026 · Sohan Patil · Ayush Ojha · Manthan Raykar · Adarsh M V · Abhinav Raj · Kushvendra Singh Rathore

START HERE

The project in one minute

> Thousands of satellites share space with millions of pieces of junk. Satellites get warned about near-misses all the time, but almost every warning is a false alarm. OrbitGuard AI predicts early which warnings will actually turn dangerous, and explains why — so people can focus on the few that matter.

1

The problem

About **40,000** objects are tracked in orbit, plus ~1.2 million smaller pieces. A satellite can get **hundreds of warnings a week**, but only about **1 in 500** is serious enough to act on. Experts must decide ~2 days ahead whether to move the satellite.

2

What we build

A free web dashboard. Pick any satellite, and it **finds every close pass in the next 3 days** using public orbit data, then uses **machine learning** trained on real ESA warnings to **predict which ones stay dangerous**, with reasons.

3

Why it matters

Fewer wasted hours on false alarms. Earlier decisions mean **smaller, cheaper moves**. One crash can create thousands of new fragments: the 2009 Iridium–Cosmos crash made **2,201**.

How it works

1. Get data
Free public orbit data + ESA's real warning data

→

2. Physics
Predict positions for 3 days, find close passes, estimate hit chance

→

3. Machine learning
Predict if the risk will stay high, explain why

→

4. Dashboard
Risk level, alert, suggested move, 3D view

An everyday analogy

Think of a smoke alarm that goes off every time you make toast. You start ignoring it — dangerous. OrbitGuard is like a smarter alarm that learns the difference between toast and a real fire, and tells you *why* it thinks it's a fire.

What we are honest about

- Our hit-chance number is a **rough estimate**, because detailed uncertainty data isn't public.
- Beating even the simplest guess is **hard**; ESA's own competition proved it. We measure ourselves against it openly.
- ESA's training data is from **2015–2019** and has names and dates hidden.

JARGON BUSTER

Every technical word, in plain English

Space debris
Junk in orbit: dead satellites, old rocket parts, and fragments from explosions and crashes. It moves at ~7.5 km per second, so even a small piece can destroy a satellite.

LEO (Low Earth Orbit)
The crowded region up to ~2,000 km above Earth. The ISS and most satellites live here.

ISS
International Space Station. Has people on board, so it dodges debris when needed.

ESA
European Space Agency — Europe's version of NASA. It published the real dataset we train on.

NASA CARA
NASA's team that checks NASA satellites for collision risk.

18th SDS
18th Space Defense Squadron — the US Space Force unit that tracks every known object in orbit and sends out collision warnings.

Satellite catalog
The official list of every tracked object in orbit. Each object gets an ID number.

Close approach / conjunction
When two objects are predicted to pass very near each other.

TCA
Time of Closest Approach — the moment the two objects are nearest.

CDM
Conjunction Data Message — a warning message saying "object X will pass near your satellite at time T". A new one arrives ~3 times a day as the event gets closer, each with an updated risk.

Pc (probability of collision)
The chance the two objects actually hit. 1 in 10,000 (written 1e-4) is the usual point where operators move the satellite.

Covariance
How unsure we are about where an object is. Bigger uncertainty → less reliable Pc. Real values are mostly not public, so we assume one.

TLE
Two-Line Element set — the free, public description of an object's orbit, in two lines of text.

SGP4
The standard formula that turns a TLE into a predicted position at any future time.

Propagate
Predict where an object will be in the future.

Manoeuvre / Δv (delta-v)
Firing a satellite's thruster to move it out of the way. Δv is how big the push is; smaller = less fuel used.

Re-entry
When an object falls back into the atmosphere and burns up.

Space weather
Activity from the Sun. Solar storms heat and puff up the upper atmosphere, which drags satellites down faster.

CelesTrak / SOCRATES
A free public website with orbit data. SOCRATES is its free close-approach list.

Space-Track
The US government website (free account) where the official catalog and basic warnings are published.

Kelvins
ESA's 2019 machine-learning competition on this exact problem, with real (anonymised) warning data from 2015–2019.

Kessler
An open-source machine-learning toolkit for collision warnings, built by Oxford, Surrey and ESA researchers. (Not to be confused with "Kessler syndrome", a chain reaction of collisions.)

Machine learning (ML)
Software that learns patterns from past examples to make predictions.

XGBoost
A popular, fast ML method that works very well on table-shaped data.

SHAP
A method that explains an ML prediction, e.g. "risk is high mainly because the miss distance keeps shrinking".

Baseline / persistence
The simplest possible guess to compare against. Persistence = "the final risk will equal the latest risk we've seen".

Recall
Of all the truly dangerous events, how many did we catch? Missing one is the worst mistake.

F2 score
A single score combining catches and false alarms, weighted so missing a dangerous event counts more than a false alarm.

Brier score
Measures whether predicted probabilities are honest (when we say 70%, does it happen ~70% of the time?). Lower is better.

False alarm
A warning that turns out harmless.

BEFORE PRESENTING

5 things to fix in the slides

These are the lines an expert judge could challenge. Everything else is backed up.

SLIDE	WHAT THE SLIDE SAYS → THE PROBLEM	VERDICT	CHANGE IT TO
3	**Kessler called a "dashboard with no forecast"** It is an ML toolkit that does forecast — the closest thing to our idea.	WRONG	Move it to the ML row. Pitch our gap as "all in one, and explained".
2	**"Fewer false alarms"** ESA found the simple guess very hard to beat (best gain ~1.3%).	NOT PROVEN YET	Say "aims to beat the simple guess" and show the real number.
3	**Agency data is "closed to public"** Basic warnings are public with a free account.	REWORD	"Detailed uncertainty data is not public."
1	**"Catalog passed ID #100,000"** It's an ID, not a count. ~40,000 objects are tracked.	REWORD	"First six-digit ID issued July 2026; ~40,000 objects tracked."
4, 6	**Checking 3 days ahead with free data** Positions drift ~12 km over 3 days.	REWORD	Keep the "rough estimate" label; mention the number in the demo.

What the verdict labels mean

TRUE	A trusted source confirms it
PROBABLY TRUE	Makes sense, but no direct proof found
REWORD	True, but the wording could mislead
NOT PROVEN YET	Don't claim it until we measure it
WRONG	A source says the opposite

01 SLIDE 1 The problem

IN SIMPLE WORDS

Satellites constantly get warnings that some piece of junk might pass close by. Almost all of these warnings turn out harmless, but someone still has to check each one — and decide about two days ahead whether to move the satellite. Our tool, OrbitGuard AI, helps sort the scary warnings from the harmless ones. The problem keeps growing because space is getting more crowded.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
Satellites get a constant stream of warnings	TRUE	ESA says a typical satellite in low orbit gets hundreds of warnings every week. About 2 per week need a human expert to look closely. Source: ESA website
Most warnings never become dangerous	TRUE	In ESA's real data of 15,321 close-approach events, only 30 (0.2%) got risky enough to normally move the satellite. ESA moves each satellite only about once a year. Source: ESA competition paper (2021), ESA website
Decisions are made about 2 days before the close approach	TRUE	ESA says moves to avoid debris are usually planned at least 2 days ahead, because preparing them takes time. Source: ESA competition paper (2021)
Object ID #100,000 was given out in July 2026	REWORD	True, but careful: it is an ID number, not a count. IDs 70,000–99,999 are reserved, so numbering jumped from 69,999 to 100,000 on 11 July 2026. The real number of tracked objects is about 40,000 (plus ~1.2 million pieces too small to track). Better wording: "First six-digit ID issued July 2026; ~40,000 objects now tracked." Source: KeepTrack, orbitalnodes, ESA report 2025
The 2009 Iridium–Cosmos crash made 2,000+ tracked pieces	TRUE	An American and a dead Russian satellite collided in 2009. NASA counted 2,201 tracked fragments, and most were still in orbit years later. Source: NASA report
Users: satellite operators, analysts, students, researchers	PROBABLY TRUE	ESA's public competition on this problem drew 96 teams, so the research community clearly cares. Big operators have their own tools, so smaller operators and students are the realistic users. Source: ESA Kelvins, ESA competition paper (2021)

TALKING POINT

Say it like this: "Out of every 500 warnings, about 1 really matters. Finding close passes is easy — knowing early which ones to worry about is hard."

02

SLIDE 2

Our solution

IN SIMPLE WORDS

Two parts work together. Part A (physics) predicts where your satellite and every tracked object will be over the next 3 days, finds close passes, and estimates the chance of a hit. Part B (machine learning) looks at the early warnings and predicts whether the risk will still be high when the moment arrives — and explains why. Part C (bonus, if time allows) estimates when an object will fall back to Earth.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
Physics finds close passes; ML predicts which stay dangerous	TRUE	This is exactly the problem ESA set in its 2019 competition: use warnings from 2+ days out to predict the final risk. So it is a real, recognised problem. Source: ESA competition paper (2021)
Helps operators focus on the warnings that matter	PROBABLY TRUE	ESA filters hundreds of weekly warnings down to about 2 that need hours of expert work each. Ranking them saves real time. Source: ESA website
Fewer false alarms	NOT PROVEN YET	Careful. In ESA's competition, the simplest guess ("final risk = latest risk") was very hard to beat. The best models later improved on it by only ~1.3%. Say "we aim to beat the simple guess" and show your real number instead. Source: ESA competition paper (2021)
Earlier warning means a smaller push (less fuel)	TRUE	A tiny push given early moves the satellite a lot by the time of the close pass. Example: with 3 days' notice, a push of about 4 mm per second is enough. Catch: predictions are also less certain that far out. Source: Dodgeball paper, Manoeuvre review (2025)
Alerts are transparent, so operators can trust them	PROBABLY TRUE	SHAP shows which factors drove each prediction. Experts in this field also ask for explainable models. Note: we don't actually test "trust". Source: Kessler paper
Part C: when an object falls depends on its history and the Sun	TRUE	In Feb 2022, a small solar storm thickened the upper air and 38 of 49 brand-new Starlink satellites fell and burned up. A study of later Starlink re-entries found that the stronger the storm, the faster they fell. Source: CIRES, Starlink re-entry study

TALKING POINT

Say it like this: "We test against the same simple guess ESA used. Beating it, even slightly, is the honest bar."

03

SLIDE 3

What exists today, and the gap

IN SIMPLE WORDS

Other tools already exist, but each has a missing piece. Military and space-agency systems have the best data but keep the details private. Free websites list close passes but don't predict or explain. Competition code uses real data but doesn't run live. Our gap: one free tool that finds close passes live, predicts which matter, and explains why.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
Military and agency teams have the most precise data	TRUE	The US Space Force unit (18th Space Defense Squadron) tracks everything in orbit and sends warnings to satellite owners. NASA says owners should move the satellite when the hit chance is above 1 in 10,000. Source: Wikipedia, NASA handbook
Their data is closed to the public	REWORD	Partly wrong. Anyone with a free Space-Track account can see basic warnings, including the hit chance. Only the detailed uncertainty data is private. Better wording: "Detailed uncertainty data is not public." Source: Space-Track API
Their risk number is a snapshot, not a forecast	PROBABLY TRUE	Operators decide using the newest warning's risk. We found no public sign that they use prediction models, though they might privately. Source: ESA competition paper (2021)
CelesTrak SOCRATES: free daily list	REWORD	Free, but it updates 3 times a day, not once. Change "daily" to "3× daily". Source: CelesTrak
SOCRATES has no ML, no explanations, no re-entry view	TRUE	Correct for SOCRATES. Note: re-entry predictions do exist on other free sites (Space-Track). Source: CelesTrak
Kelvins competition code uses real ESA data, offline only	TRUE	ESA's dataset: 15,321 events, ~199,000 warnings, 2015–2019, with names and dates hidden. Source: ESA competition paper (2021)
Competition code is rarely compared to a simple baseline	NOT PROVEN YET	We didn't check this, and ESA's competition did compare everyone to a simple baseline. Safer: "rarely combined with live screening".
KESSLER is a dashboard with no forecast	WRONG	Kessler is not a dashboard. It is a machine-learning toolkit that does forecast how warnings change — the closest thing to our idea. Move it to the ML row and name a real dashboard (e.g. KeepTrack) instead. Source: Kessler paper
OrbitGuard combines everything in one tool	PROBABLY TRUE	We found no free tool that does live screening, ML prediction and explanations together. Pitch the gap as "all in one, and explained", not "nobody has done ML".

04

SLIDE 4

How the system works

IN SIMPLE WORDS

Free orbit data is downloaded and saved locally. Then two engines run side by side: the physics engine predicts positions and finds close passes; the ML engine scores the risk and explains it. A decision step turns this into a risk level, an alert, and a suggested push size. Everything shows up in a simple web dashboard. User flow: pick a satellite → check next 3 days → open the riskiest event → see risk + reasons → get alert and suggested move.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
Our data sources are free and usable	TRUE	CelesTrak is open; Space-Track needs a free account; ESA released its competition data publicly in 2019. Source: CelesTrak, ESA competition paper (2021)
SGP4 is the right formula for public orbit data	TRUE	Public orbit data (TLEs) is designed to be used with SGP4. A 2026 study found fancier methods did no better with the same data. Source: SGP4 paper (2006), Starlink accuracy study (2026)
Our hit-chance number is rough, not mission-grade	TRUE	NASA says orbit data without uncertainty info makes risk numbers less accurate. The free SOCRATES site has the same limit. Being upfront about it is correct. Source: NASA handbook, CelesTrak
Checking 3 days ahead	REWORD	Works, but public orbit data drifts: about 1 km off at first, about 12 km off after 3 days. Say this number in the demo so judges know you understand it. Source: Starlink accuracy study (2026)
Using solar activity as an input	TRUE	Solar storms clearly change how fast objects fall. Storm strength matters more than the Sun's general brightness. Source: Starlink re-entry study
Suggesting a push size	PROBABLY TRUE	Typical avoidance pushes are tiny (around 1 cm per second). Show ours as a rough estimate, not a real plan. Source: Manoeuvre review (2025)

TO DO

The slide still says "Architecture diagram: to be inserted by the team". Add the diagram before submitting.

05

SLIDE 5

Tools and 12-hour plan

IN SIMPLE WORDS

Everything is written in Python and runs on one normal laptop — no special hardware or paid cloud. Streamlit makes the web dashboard, Plotly draws the 3D orbits, and standard science libraries do the maths. The ML uses XGBoost plus SHAP for explanations. Four team members each own one part, over a 12-hour plan.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
SGP4 is the standard	REWORD	Standard for free public data. Professional teams use more precise private methods. Say “standard for public orbit data”. Source: SGP4 paper (2006)
XGBoost is strong on table data	TRUE	A 2022 study across 45 datasets found this type of method beats deep learning on table-shaped data. Our warnings are tables (103 columns each). Source: NeurIPS 2022 study
This type of model works on this exact data	TRUE	After ESA’s competition, a very similar method (LightGBM) gave the only clear improvement over the simple guess. Source: ESA competition paper (2021)
SHAP explains each prediction	TRUE	SHAP is a widely used, well-tested explanation method for exactly this kind of model (published 2017 and 2020).
Runs on one laptop, no GPU	PROBABLY TRUE	~199,000 rows trains in minutes on a normal laptop. Checking one satellite against ~30,000 objects is fine; checking everything against everything is not — which is why that’s listed as future work.
Free data, zero setup	REWORD	Free, but Space-Track limits downloads (under 30 per minute, and only 3 full warning downloads a day). Download and save everything in the first hour. Source: Space-Track limits, Space-Track API

TO DO

The plan says “freeze at noon” but never says when the 12 hours start. Add the start time.

06

SLIDE 6

Results, demo and risks

IN SIMPLE WORDS

By the end we show: a working dashboard, the ML model compared against the simple guess, a page of real scores, and saved demo data in case the internet fails. The 3-minute demo: pick the ISS → check the next 3 days → view the riskiest event in 3D → replay a real ESA event → show scores.

WHAT WE CLAIM	VERDICT	PROOF / EXPLANATION
Demo on the ISS	TRUE	Great choice: the ISS has dodged debris 39 times since 1998, most recently in November 2024. Source: Live Science
Replay a real ESA event	REWORD	ESA hid names, dates and real positions in its data. Call it "a real anonymised ESA event" and don't draw it on the ISS orbit. Source: ESA competition paper (2021)
Main score: recall / F2	TRUE	ESA used F2 for the same reason: missing a real danger is worse than a false alarm. Use ESA's risk cut-off so you can compare to the simple guess's F2 of 0.739. Source: ESA competition paper (2021)
Brier score	TRUE	The standard way to check whether predicted chances are honest (since 1950).
Comparing against the simple guess	TRUE	ESA's simple-guess numbers to beat: F2 = 0.739 on its official test set. Source: ESA competition paper (2021)
Risk: download limits	TRUE	Confirmed: under 30 downloads a minute, 300 an hour, 3 full warning lists a day. Source: Space-Track limits, Space-Track API
Risk: ESA data is old (2015–2019) and anonymised	TRUE	Confirmed. Also, only ~200 truly risky events exist in the test data, so scores will jump around. Source: ESA competition paper (2021)
Future ideas	PROBABLY TRUE	Fast GPU orbit prediction and better models of the Sun's effect are both active research areas. Source: GPU study, Starlink re-entry study

TO DO

Add one more risk: "our model may not beat the simple guess." ESA found many models looked good in practice but failed on the real test. Say what you'll show if that happens (the explanations still help).

PROOF

Sources

Start with the first one: ESA's write-up of the exact problem we are solving.

1. ESA competition paper — Uriot et al., "Spacecraft Collision Avoidance Challenge", 2021
   https://arxiv.org/pdf/2008.03069
2. ESA — Automating collision avoidance
   https://www.esa.int/Space_Safety/Space_Debris/Automating_collision_avoidance
3. ESA Kelvins — Collision Avoidance Challenge
   https://kelvins.esa.int/collision-avoidance-challenge/
4. ESA Space Environment Report 2025
   https://www.esa.int/Space_Safety/Space_Debris/ESA_Space_Environment_Report_2025
5. KeepTrack — Object 100000
   https://keeptrack.space/deep-dive/satellite-100000-saramago
6. orbitalnodes — Satellite catalog numbers
   https://orbitalnodes.ai/satellite-catalog-numbers/
7. NASA — Iridium 33 / Cosmos 2251 debris
   https://ntrs.nasa.gov/api/citations/20150003820/downloads/20150003820.pdf
8. NASA Collision Avoidance Best Practices Handbook
   https://www.nasa.gov/wp-content/uploads/2021/07/nasa_ca_handbook_for_s3vi_20211013.pdf
9. 18th Space Defense Squadron — Wikipedia
   https://en.wikipedia.org/wiki/18th_Space_Defense_Squadron
10. CelesTrak SOCRATES
    https://celestrak.org/SOCRATES
11. Space-Track API listing
    https://apis.io/apis/spacetrack/spacetrack-conjunction-data-api/
12. Space-Track download limits
    https://spacetrack.readthedocs.io/en/stable/usage.html
13. Acciarini et al., "Kessler: an ML library for spacecraft collision avoidance"
    https://oatml.cs.ox.ac.uk/publications/202101_AcciariniAutomated.html
14. Vallado et al., "Revisiting Spacetrack Report #3" (SGP4), 2006
    https://celestrak.org/publications/AIAA/2006-6753/
15. "How long can you trust a Starlink TLE?", 2026
    https://arxiv.org/pdf/2605.19850
16. "Dodgeball — Can a Satellite Avoid Being Hit by Debris?"
    https://arxiv.org/pdf/2110.03113
17. "CAMmary: Review of Collision Avoidance Manoeuvre Design", 2025
    https://arxiv.org/pdf/2503.22555
18. CIRES — Feb 2022 Starlink loss
    https://cires.colorado.edu/news/minor-geomagnetic-storm-big-impact-february-2022-starlink-satellite-loss
19. "Tracking Reentries of Starlink Satellites", 2025
    https://arxiv.org/pdf/2505.13752
20. Grinsztajn et al., "Why do tree-based models still outperform deep learning on tabular data?", NeurIPS 2022
    https://arxiv.org/abs/2207.08815v1
21. Live Science — ISS dodges its 39th piece of space junk
    https://livescience.com/space/space-exploration/iss-dodges-its-39th-piece-of-potentially-hazardous-space-junk-experts-say-it-wont-be-the-last
22. jaxsgp4 — GPU orbit propagation, 2026 (title only)
    https://arxiv.org/pdf/2603.27830

Also used: Lundberg & Lee 2017 and Lundberg et al. 2020 (SHAP); Brier 1950 (Brier score).